

Agile Testing



Day 13



01 Overview of Agile

Client and management **myths**

If you work with a non-Agile aware executive management or client, their expectations will be:

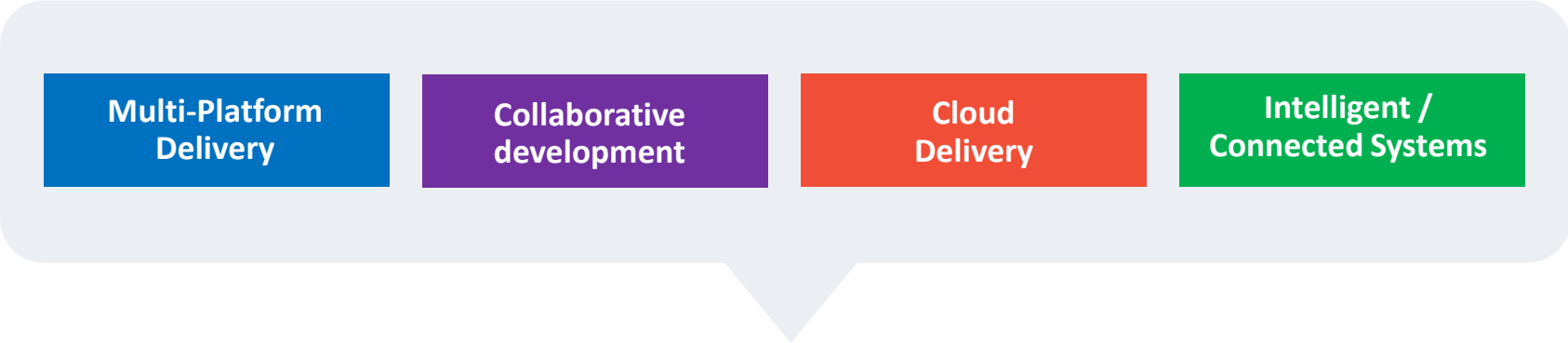
- You will accept change continuously
- You will suddenly become more productive
- You are doing development without any discipline

© Copyright IBM Corporation 2015

Why **Agile**?

Brighter Blue

Software development is to be adopted for new ways of working and technology changes.



Impacts on Business and Software Development:

- Heavy demand for rapid product lifecycles
- Increasing interactions with customer
- Exploring adjacent or new markets
- New skills and competencies needed
- Increased security and privacy requirements
- Greater need to align with strategic priorities

© Copyright IBM Corporation 2015

The Agile **manifesto – 2001**

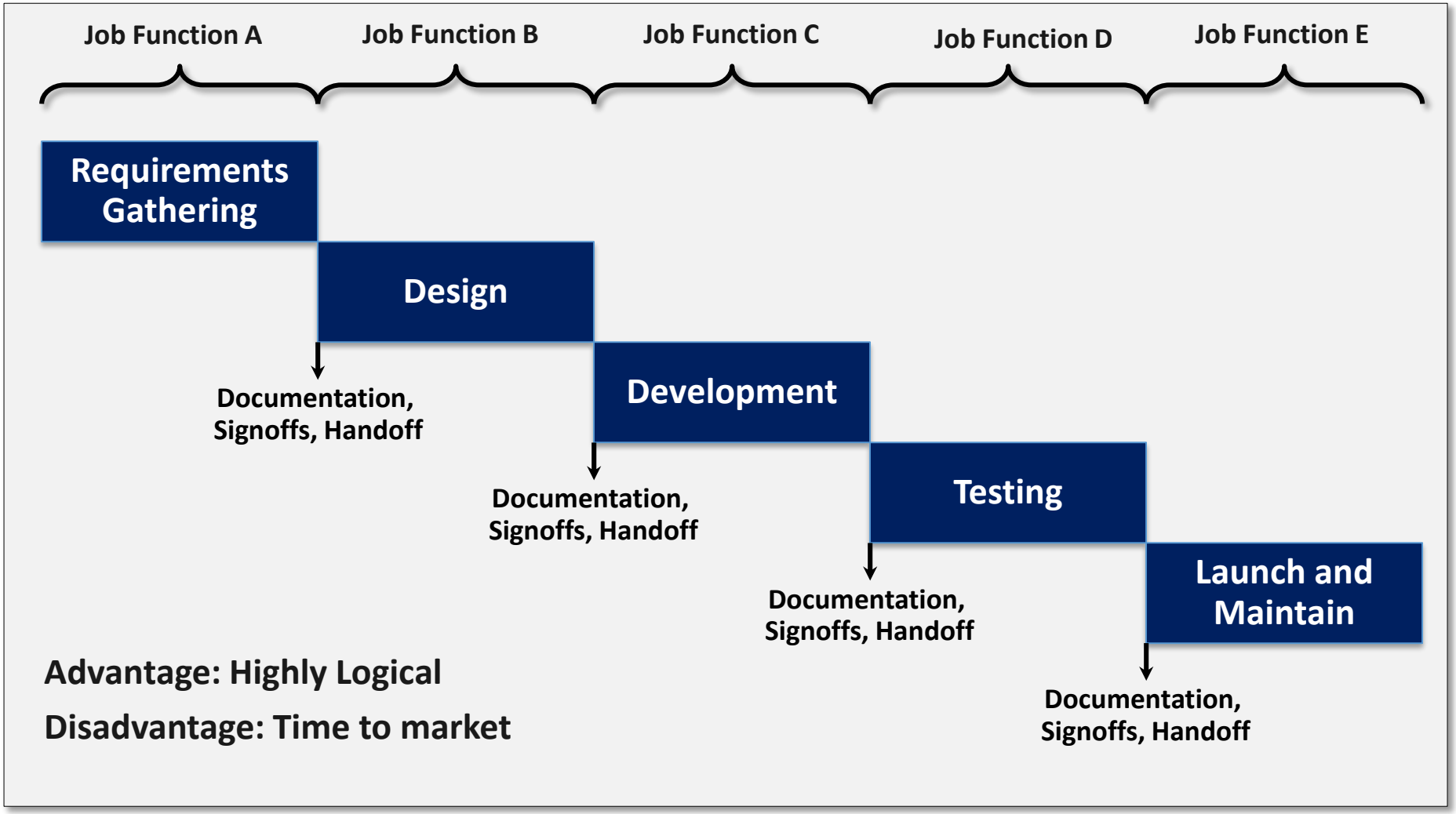
We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions	>	processes and tools
Working software	>	comprehensive documentation
Customer collaboration	>	contract negotiation
Responding to change	>	following a plan

It means that while there is value in the items on the right, **we value the items on the left more**.

© Copyright IBM Corporation 2015

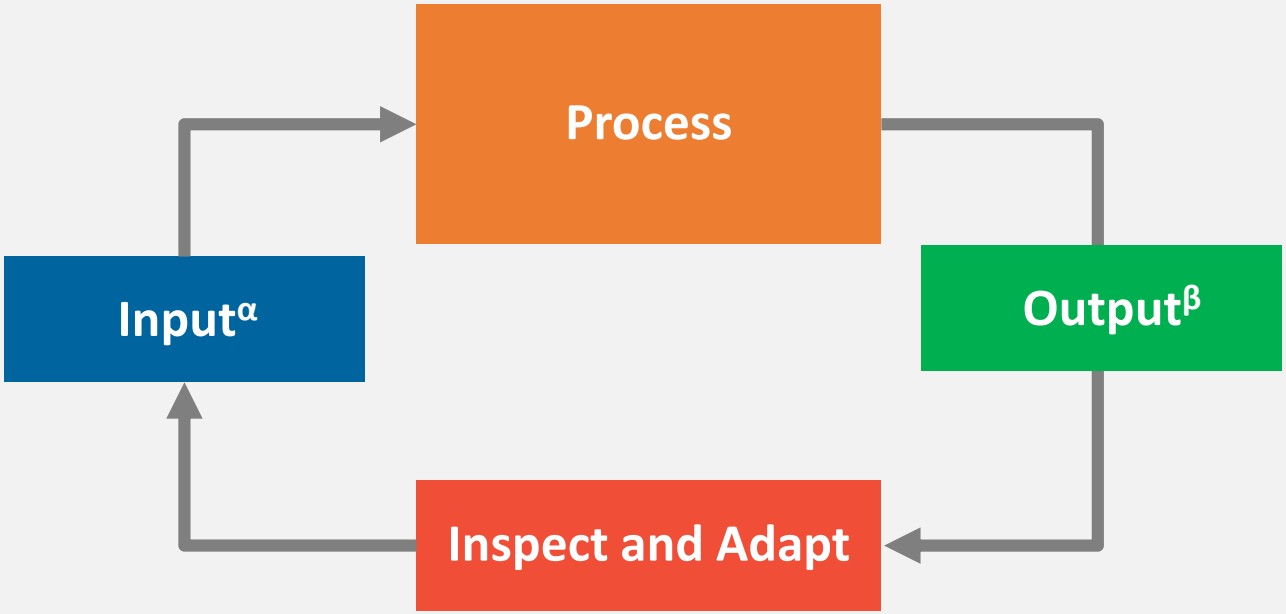
Waterfall model



© Copyright IBM Corporation 2015

Empirical processes

The Empirical process is an inspect and adapt process where the product is built incrementally.



© Copyright IBM Corporation 2015

Brighter Blue

02 Scrum Basics

IBM

What is Scrum?

Scrum refers to a holistic or “rugby” approach—where teams go the distance as a unit, passing the ball back and forth—as opposed to the traditional sequential or “relay race” approach for managing new product development.

It is not individual effort but the best coordination of the team that will determine success.

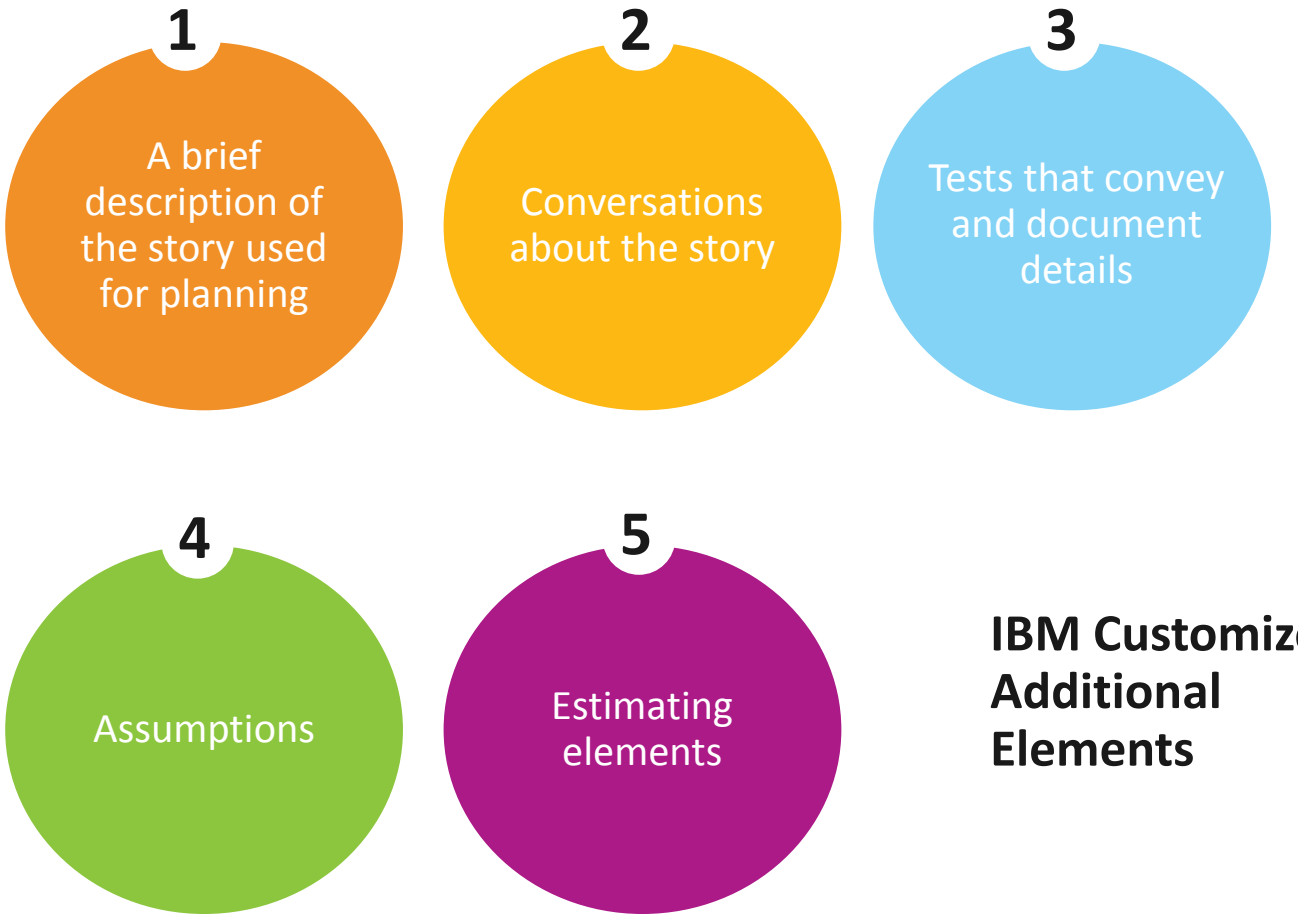


© Copyright IBM Corporation 2015

User Stories: **Components**

A User Story describes functionality that will be useful to a stakeholder of the system.

User Stories consists of three things



IBM Customized Additional Elements

© Copyright IBM Corporation 2015

An example of user stories

Brighter Blue

User Story

Success would be …

- Acceptance Test #1
- Acceptance Test #2

As a system admin at Wal*Mart, I want to configure WAS so I can manage my applications

Success would be …

- WAS is configured
- No critical errors occurred in the process
- No data was lost

© Copyright IBM Corporation 2015

What is an Epic?

Brighter Blue

EPIC

As an user, I want to cancel a reservation so that I am not charged for a travel arrangement I no longer need.

User Story

As a premium site member, I want to cancel a reservation on the last minute with no cancellation fee so that I am not charged for a travel arrangement I no longer need.

As a non-premium member, I want to cancel 24 hours in advance so that I am not charged for a travel arrangement I no longer need.

As a site visitor,
I want to receive a confirmation of any cancelled reservation so that I can have proof of cancellations.

© Copyright IBM Corporation 2015

Product Backlog

Brighter Blue

To-Do List

ID	Story	Estimation	Priority
7	As an unauthorized user, I want to create a New account	3	1
1	As an unauthorized user, I want to login	1	2
10	As an authorized user, I want to logout	1	3
9	Create script to purge database	1	4
2	As an authorized user, I want to see the list of items so that I can select one	2	5
4	As an authorized user, I want to add a new item so that it appears in the list	5	6
3	As an authorized user, I want to delete the selected item	2	7
5	As an authorized user, I want to edit the selected item	5	8
6	As an authorized user, I want to set a reminder for a selected item so that I am reminded when item is due	8	9
8	As an administrator, I want to see the list of accounts on login	2	10
Total		30	

© Copyright IBM Corporation 2015

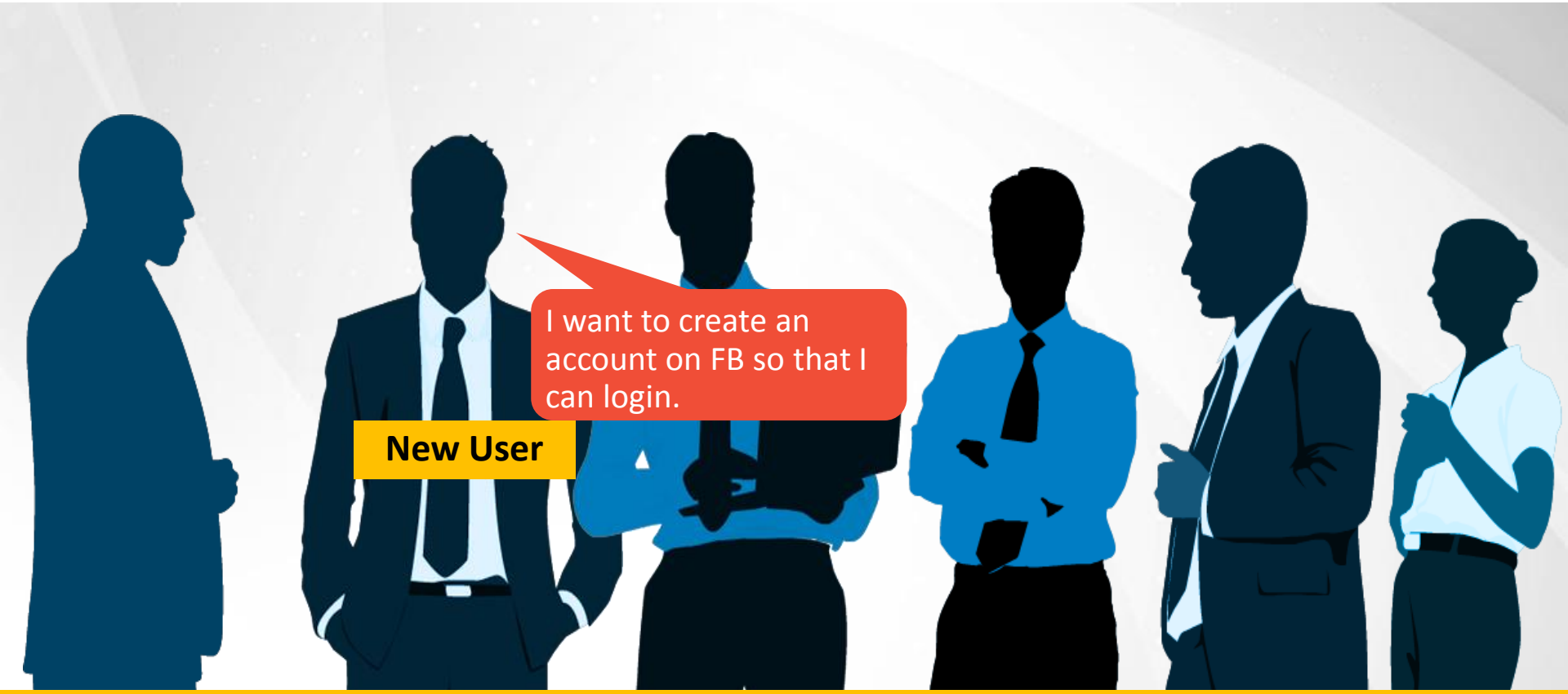
Case study: Some possible user stories (1 of 6)



Suppose the Vice President of Facebook (FB) requires a portal for FB. What could be some possible user stories?

© Copyright IBM Corporation 2015

Case study: Some possible user stories (2 of 6)



© Copyright IBM Corporation 2015

Case study: Some possible user stories (3 of 6)



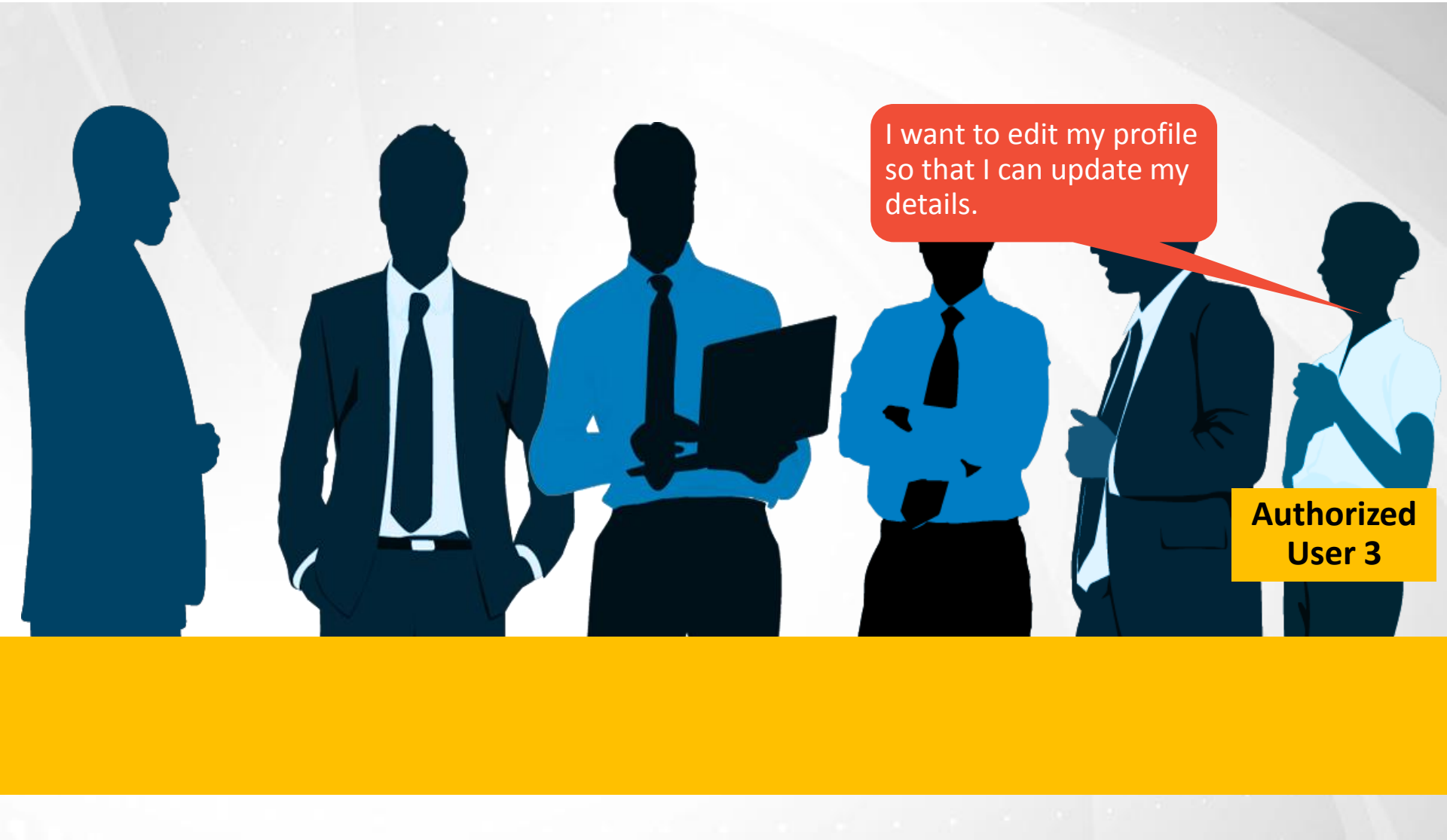
© Copyright IBM Corporation 2015

Case study: Some possible user stories (4 of 6)



© Copyright IBM Corporation 2015

Case study: Some possible user stories (5 of 6)



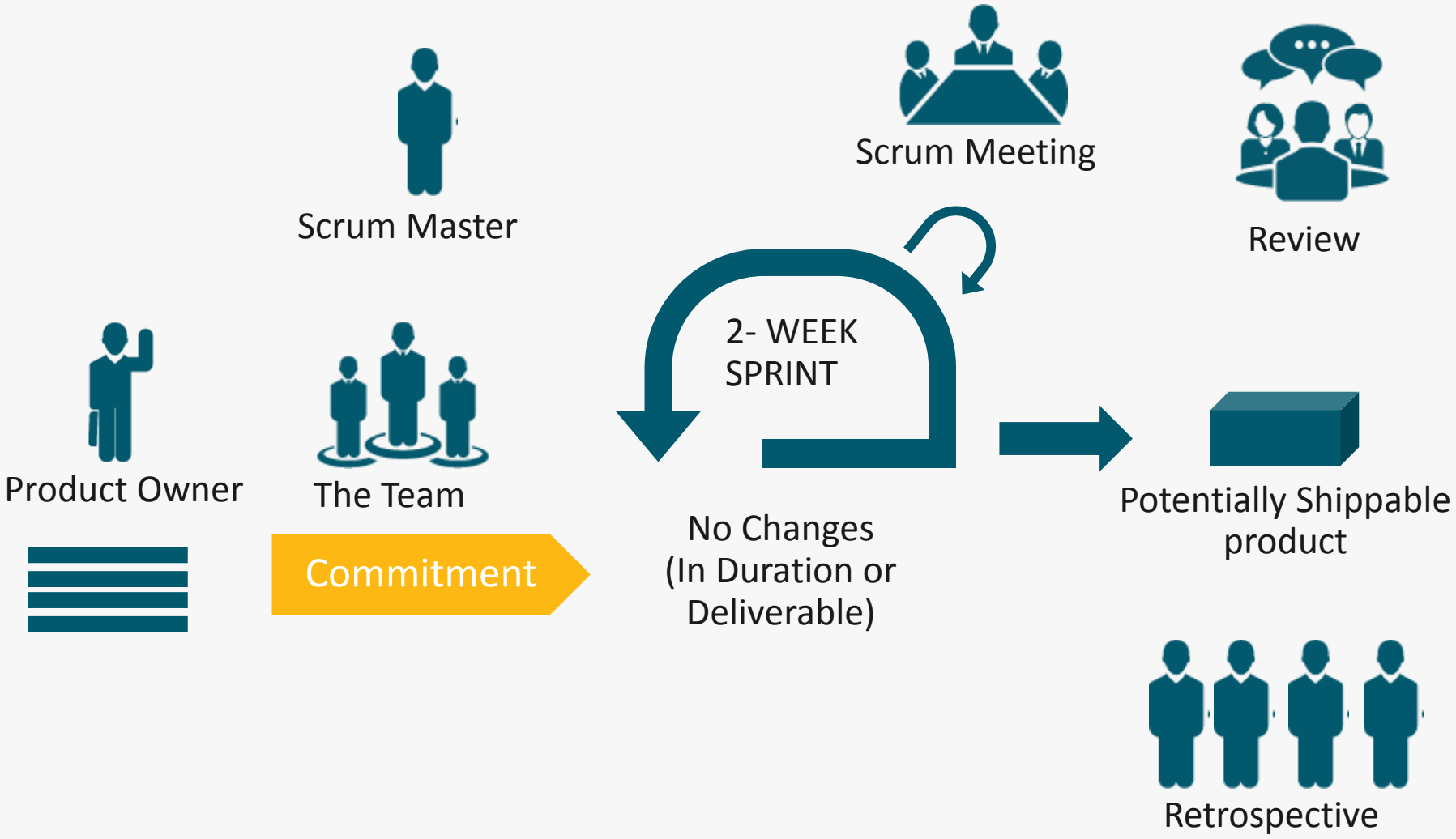
© Copyright IBM Corporation 2015

Case study: Some possible user stories (6 of 6)



© Copyright IBM Corporation 2015

Scrum roles and responsibilities



© Copyright IBM Corporation 2015

Scrum roles and responsibilities—**product owner**



© Copyright IBM Corporation 2015

Scrum roles and responsibilities—the team

7 people + or – 2

- Can be shared with other teams (but it is better when not)
- Can change between Sprints (but better when they don't)
- Can be distributed (but better when collocated)

Cross-functional

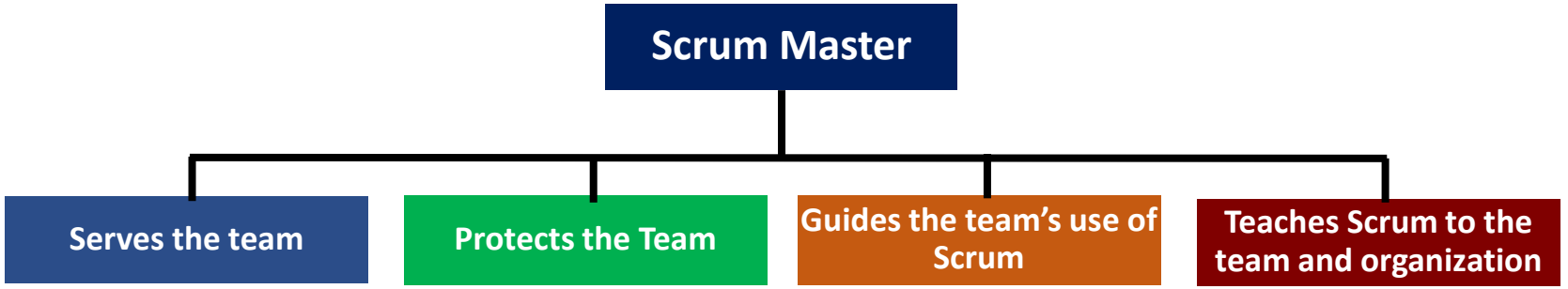
- Possesses all the skills necessary to produce an increment of potentially shippable product
- Team takes on tasks based on skills, not just the official "role".

Self-managing

- Team manages itself to achieve the Sprint commitment.

© Copyright IBM Corporation 2015

Scrum roles and responsibilities—**Scrum Master**



The Scrum Master takes action to help remove impediments to the team's effectiveness.

The Scrum Master facilitates the team's group interactions to help the team achieve its full potential.

The Scrum Master protects the team from anything that threatens its effectiveness, such as outside interference or disruption.

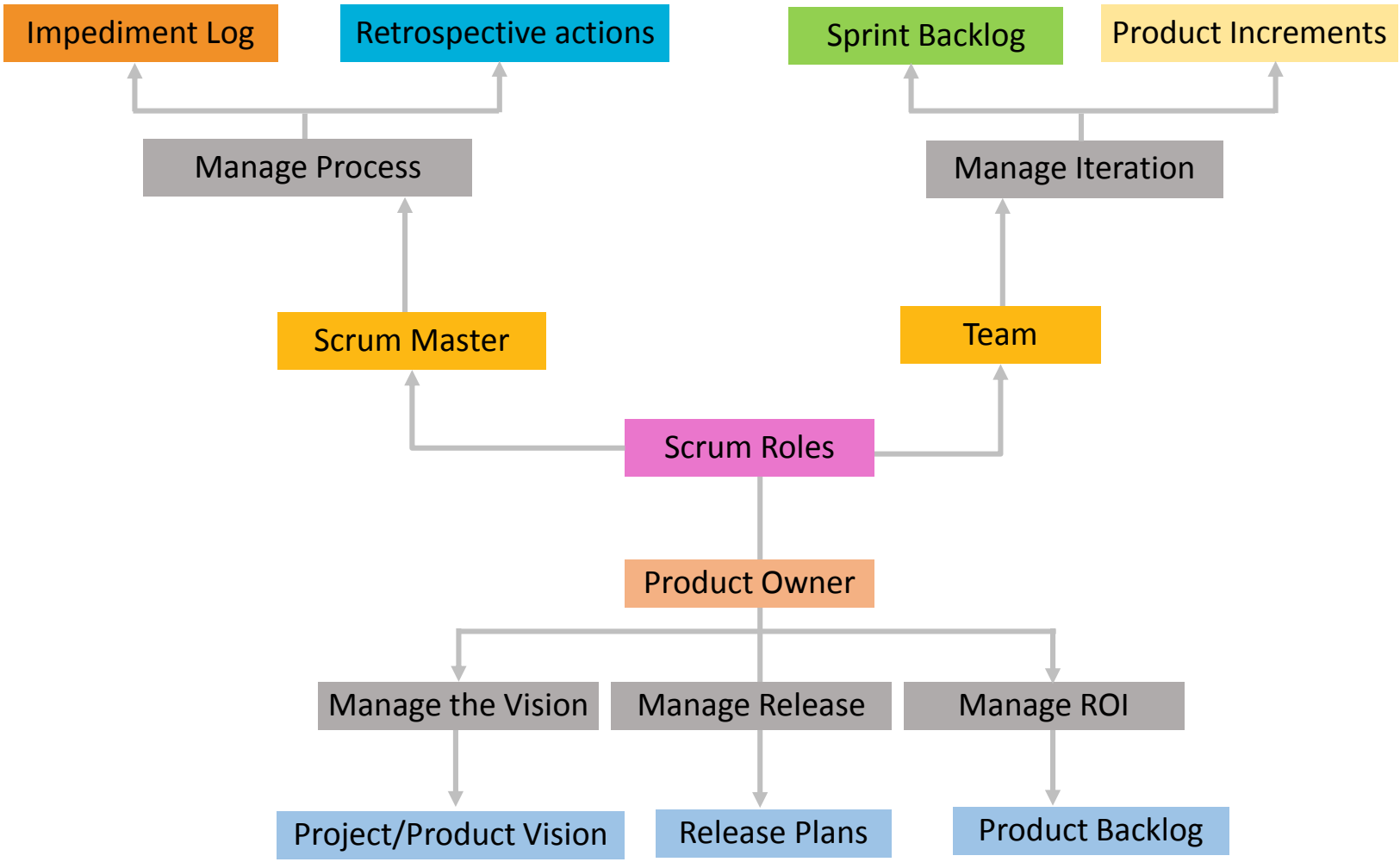
The Scrum Master will need to confront uncomfortable issues, both inside and outside the team.

The Scrum Master ensures that all standard Scrum rules and practices are followed.

The Scrum Master organizes all Scrum-related practices.

© Copyright IBM Corporation 2015

Scrum **project roles** in a nutshell



© Copyright IBM Corporation 2015

Sprint **planning**

Brighter Blue

What is on the Product Owner's "shopping list"?

Team understands details of what Product Owner has prioritized on Product Backlog.

How much "money" do we have in our bank account?

Team decides how much productive time is available during the Sprint.

How many items on the shopping list can we afford to "buy" with that "money"?

Team decides how many Product Backlog items can be committed for completion during the Sprint.

© Copyright IBM Corporation 2015

Sprint Backlog

Brighter Blue

			Day of Sprint						
Backlog Item	**Task**	**Owner**	**Initial Est.**	**1**	**2**	**3**	**4**	**5**	**6**
Enable all users to place book in shopping cart	Design business logic	Sanjay	4						
	Design user interface	Jing	2						
	Implement back-end code	Tracy	2						
	Implement front-end code	Tracy	6						
	Complete documentation	Joe	8						
	Unit testing	Philip	4						
	Regression testing	Philip	2						
Upgrade transaction processing module	Implement back-end code	Tracy	5						
	Complete documentation	Joe	6						
	Unit testing	Philip	3						
	Regression testing	Philip	3						
Total			214						

© Copyright IBM Corporation 2015
IBM

Sprint Cycle: 2 Weeks

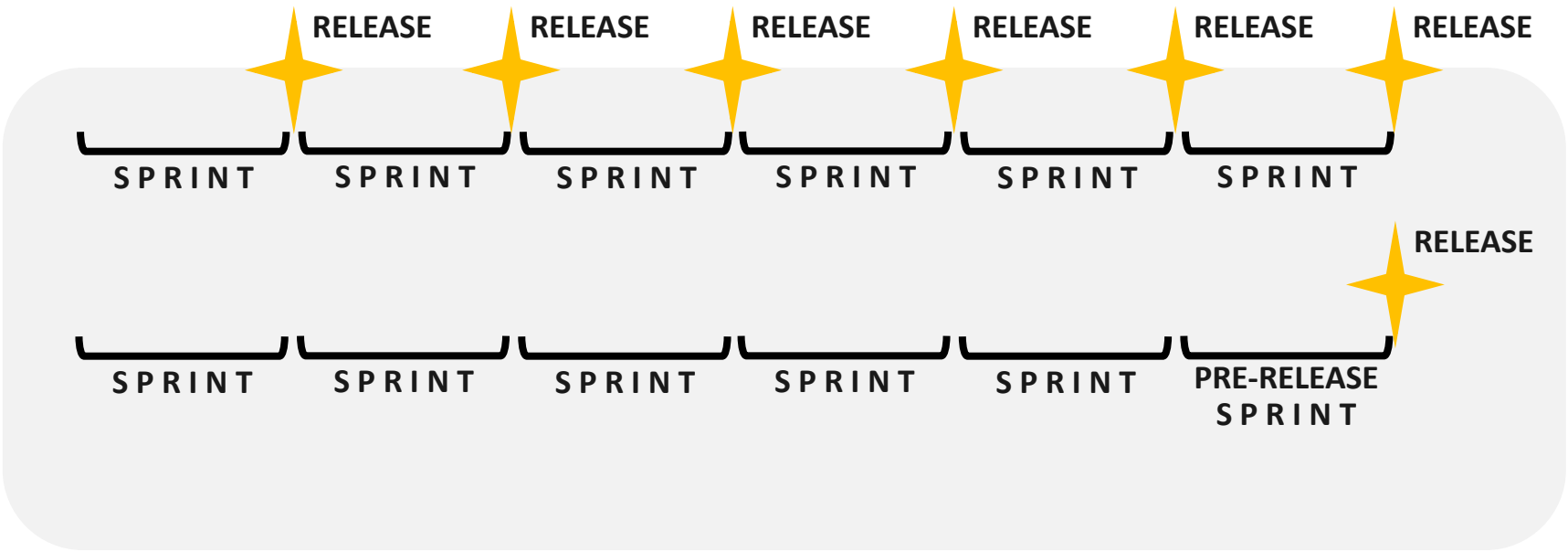
Brighter Blue

Mon	Tues	Weds	Thurs	Fri
5	6	1 **1-Hr Pre-Meeting for next Sprint**	2 8	3 **Sprint Review & Retrospective** 9
6 **Sprint Planning Meeting**	7 1	8 2	9 3	10 4
13 5	14 6	15 **1-Hr Pre-Meeting for next Sprint** 2	16 8	17 **Sprint Review & Retrospective** 4
20 **Sprint Planning Meeting**	21 1	22 2	23 3	24 4
27 5	28 6	29 **1-Hr Pre-Meeting for next Sprint**	30 8	31 **Sprint Review & Retrospective** 9

© Copyright IBM Corporation 2015

Definition of Done

In Scrum, “Done” is defined as “**Potentially Shippable**”.



© Copyright IBM Corporation 2015



© Copyright IBM Corporation 2015

Spot Quiz Sample

Brighter Blue

01 Which one of these life cycle models does Agile follow?

A Waterfall Model

B Iterative and Incremental Model

C V Model

© Copyright IBM Corporation 2015

Spot Quiz

Brighter Blue

02

Which of the following prioritized items does the product backlog contain?

A Epics

B User Stories

C Both A & B

© Copyright IBM Corporation 2015

Spot Quiz

Brighter Blue

03

Which of the below roles is not defined by Scrum?

A	Product Owner
B	Project Manager
C	Developer / Tester
D	Scrum Master

© Copyright IBM Corporation 2015

Brighter Blue

03 Scrum Execution

IBM

Daily Scrum Meeting

Brighter Blue

- Every weekday
- Whole team attends
- Everyone stands
- Lasts 15 minutes or less
- Product Owner can attend and report
- Update of artifacts after standup—Sprint backlog, Task board, and so on

© Copyright IBM Corporation 2015

Daily Scrum Meeting (continued)

Every Developer / Tester reports the following three things:

- What task was I able to accomplish since yesterday?
- What task will I try to accomplish for the day?
- Any issues blocking my work?

No discussions or conversations until meeting ends

© Copyright IBM Corporation 2015

Activity



Brighter Blue

Conduct Daily Scrum Meeting in the following scenario:

There are three developers and two testers working on building a product that will automate the Bank Transactions.
One person is working on Creating new accounts.

Sub-tasks:

1. Create a Registration/login page.
2. Add options for User to select Creating new account.
3. Add all the mandatory / optional fields as text box or combo boxes.
4. Validate the data entered and the formats and save the data in the database.
5. Generate the account number and display.

One person is working on Creating Fixed Deposit.

Sub-tasks

1. Provide the user list of FD plans and the rates.
2. Check if User is already an account holder. If yes, then display a webpage which will get the User account details so that FD account to be linked with S/B account.

One person is working on Generating Reports.

Sub-tasks:

1. Design a screen/page to provide the Report generating options.
2. Create the tables and views.
3. Design the report layout for different queries (ex: fetch all the details of Customer XYZ like account summary, FD details, Home loan details and so on.

2 Testers testing first 2 scenarios.

© Copyright IBM Corporation 2015

Sprint Backlog Updation

			Day of Sprint						
Backlog Item	**Task**	**Owner**	**Initial Est.**	**1**	**2**	**3**	**4**	**5**	**6**
Enable all users to place book in shopping cart	Design business logic	Sanjay	4	2	0				
	Design user interface	Jing	2	2	2				
	Implement back-end code	Tracy	2	4	2				
	Implement front-end code	Tracy	6	6	6				
	Complete documentation	Joe	8	6	6				
	Unit testing	Philip	4	3	3				
	Regression testing	Philip	2	2	2				
Upgrade transaction processing module	Implement back-end code	Tracy	5	10	8				
	Complete documentation	Joe	6	6	6				
	Unit testing	Philip	3	3	3				
	Regression testing	Philip	3	2	2				
Total			**214**	**220**	**205**				

© Copyright IBM Corporation 2015

Kanban Task Board (1 of 4)

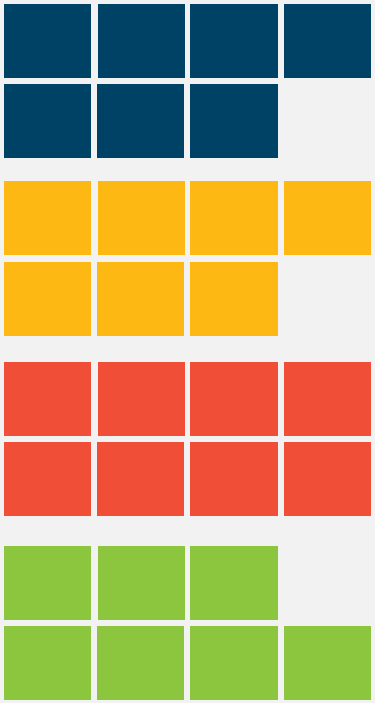
A Kanban board is a work and workflow visualization tool that enables you to optimize the flow of your work. Physical Kanban boards, like the one pictured below, typically use sticky notes on a whiteboard to communicate status, progress, and issues.



© Copyright IBM Corporation 2015

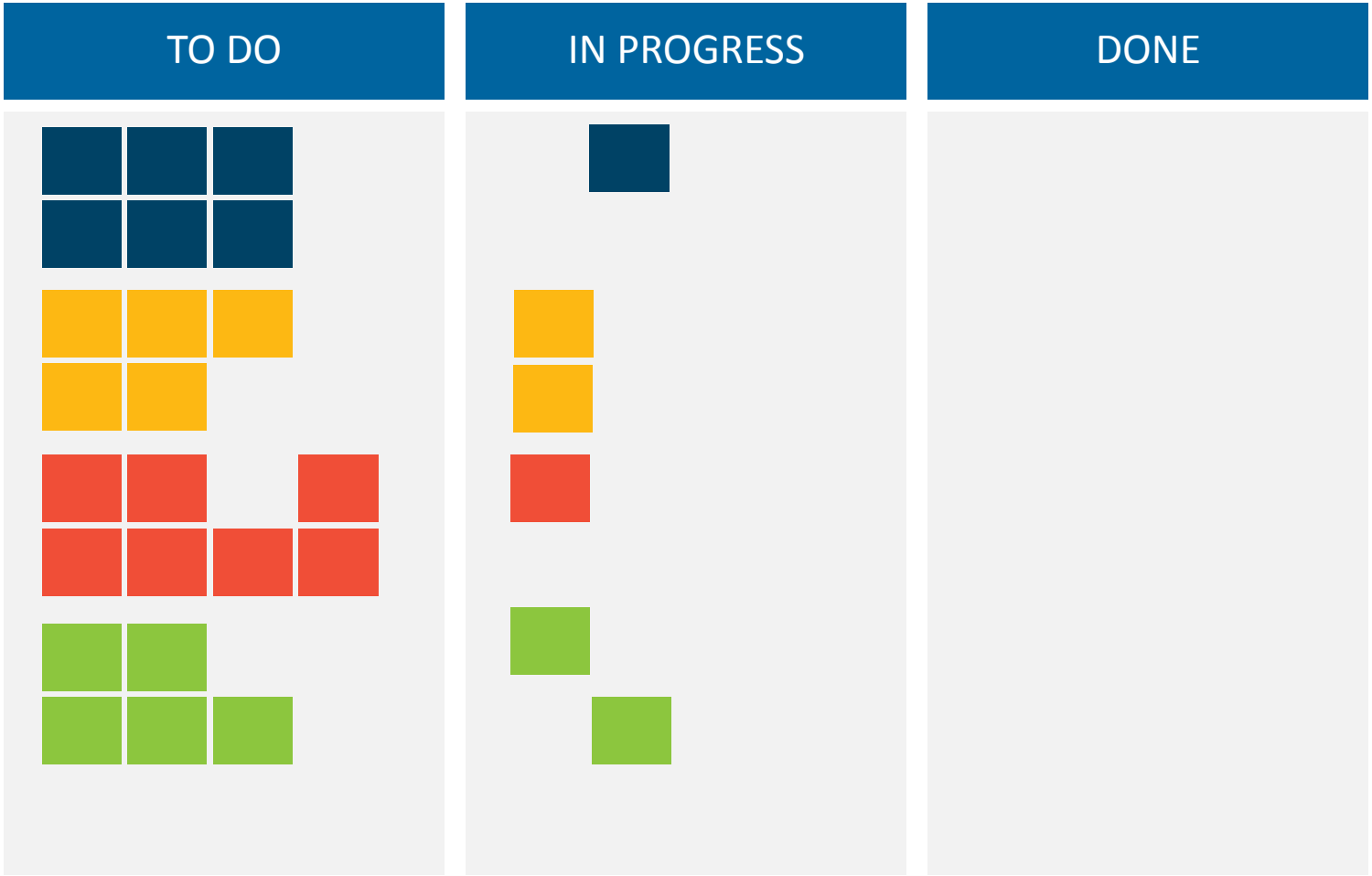
Kanban Task Board (2 of 4)

Brighter Blue

TO DO	IN PROGRESS	DONE
		

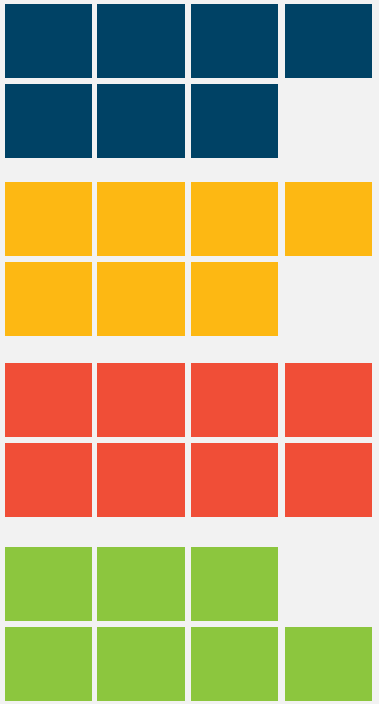
© Copyright IBM Corporation 2015

Kanban Task Board (3 of 4)



© Copyright IBM Corporation 2015

Kanban Task Board (4 of 4)

TO DO	IN PROGRESS	DONE
		

© Copyright IBM Corporation 2015



© Copyright IBM Corporation 2015

Spot Quiz Sample

Brighter Blue

01 Product Owner has to mandatorily participate in Daily Scrum Meeting.

A	Yes 	B	No
C	Not required	D	Need basis

© Copyright IBM Corporation 2015

Spot Quiz Sample

Brighter Blue

02

Sprint backlog is updated by:

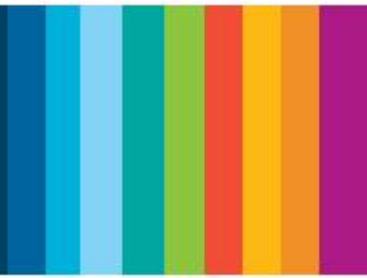
A	Scrum Master
B	Scrum Team
C	Scrum Product Owner
D	Any of the above



© Copyright IBM Corporation 2015

Brighter Blue

05 Sprint Review and Retrospective



IBM

Sprint Review

- The purpose of the Sprint Review is:
  - Demo what the team has built
  - Generate feedback which the Product Owner can incorporate in the Product Backlog
- Attended by Team, Product Owner, ScrumMaster, functional managers, and any other stakeholders
- A demo of what's been built and not a presentation about what's been built:
  - No Power Points allowed!
- Usually lasts 1-2 hours
- Followed by Sprint Retrospective

© Copyright IBM Corporation 2015

Sprint Retrospective

Brighter Blue

What is it?

- 1-2 hour meeting following each Sprint Demo
- Attended by Product Owner, Team, and ScrumMaster
- Usually a neutral person will be invited in to facilitate
- Presents what is working and what could work better

Why does the Retrospective matter?

- Accelerates visibility
- Accelerates action to improve

© Copyright IBM Corporation 2015

Sprint Retrospective: Four Square Method

Brighter Blue

A look at the Four Square Method

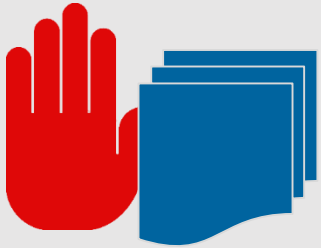
Went Well



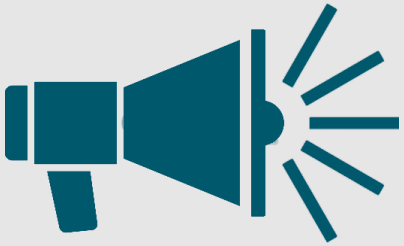
Need Improvements



Stop



Shout Out



© Copyright IBM Corporation 2015



© Copyright IBM Corporation 2015

Spot Quiz Sample

Brighter Blue

01 Which practice does not fall under technical best practices?

A	Automated build and Continuous Integration
B	Automated Regression Testing
C	Code Review and Rework
D	None of the above

© Copyright IBM Corporation 2015

Spot Quiz Sample

Brighter Blue

02 Retrospective meetings help us to:

A	Identify areas of improvement 
B	Recognize team members
C	Inspect and Adapt
D	A & B Only

© Copyright IBM Corporation 2015

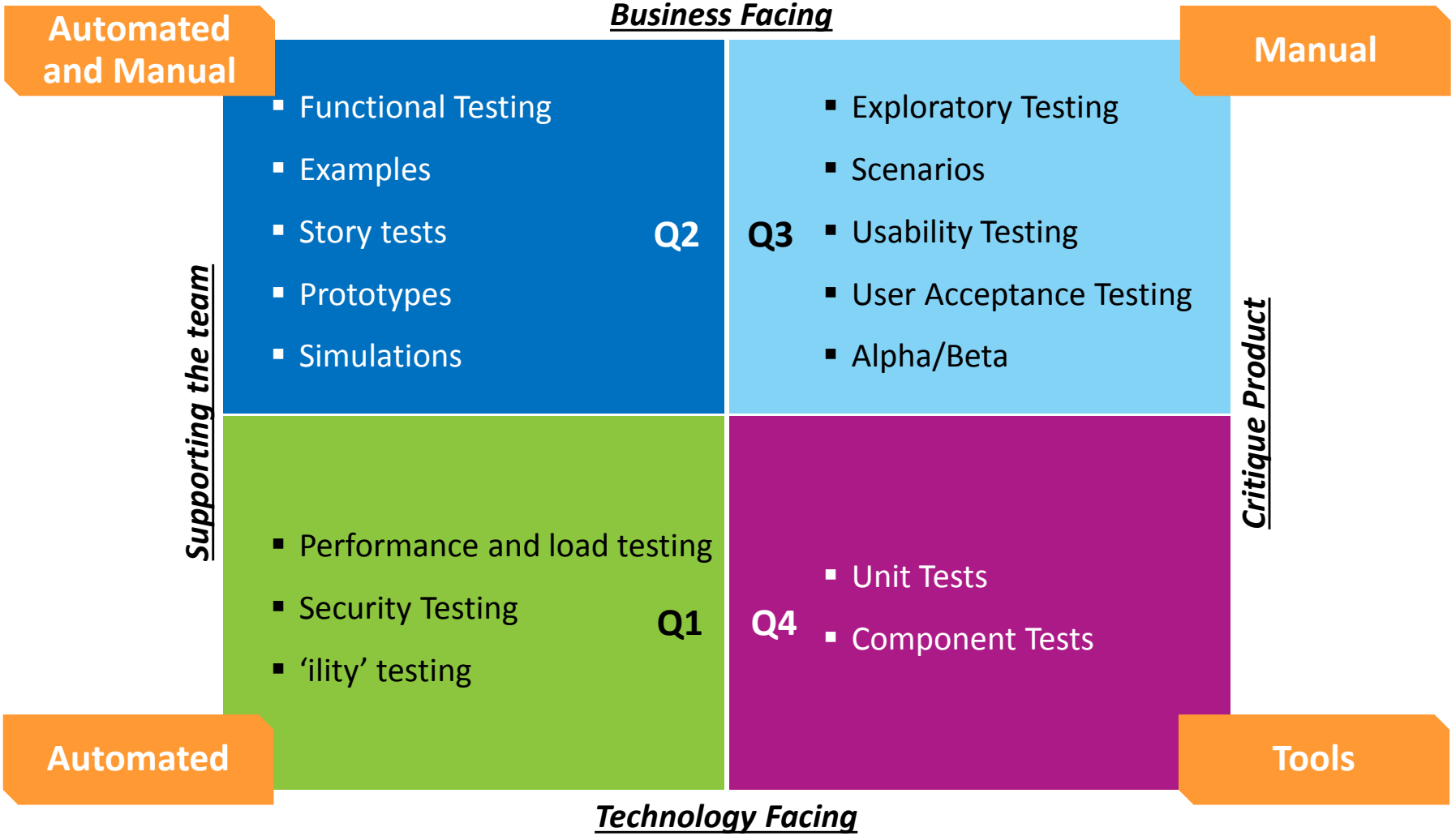
Brighter Blue

06 Agile - Testing

IBM

Agile Test Quadrants

Agile testing covers all types of testing. The Agile Testing Quadrants provide a helpful taxonomy to help teams identify and plan the testing needed.



© Copyright IBM Corporation 2015

Agile Test Quadrants (continued)

Brighter Blue

Quadrant 1:

- Depicts unit and component tests written by developers
- Largely automated and represent a low cost of ensuring that the system works as intended

Quadrant 2:

- Depicts user facing story and functional tests
- Tests the higher level features of the system and can be both automated and manual

Quadrant 3:

- Depicts system-level tests to determine the aggregate behavior of the system meets its usability and functionality requirements
- Largely manual since they involve users and testers using the system in actual or simulated deployment

Quadrant 4:

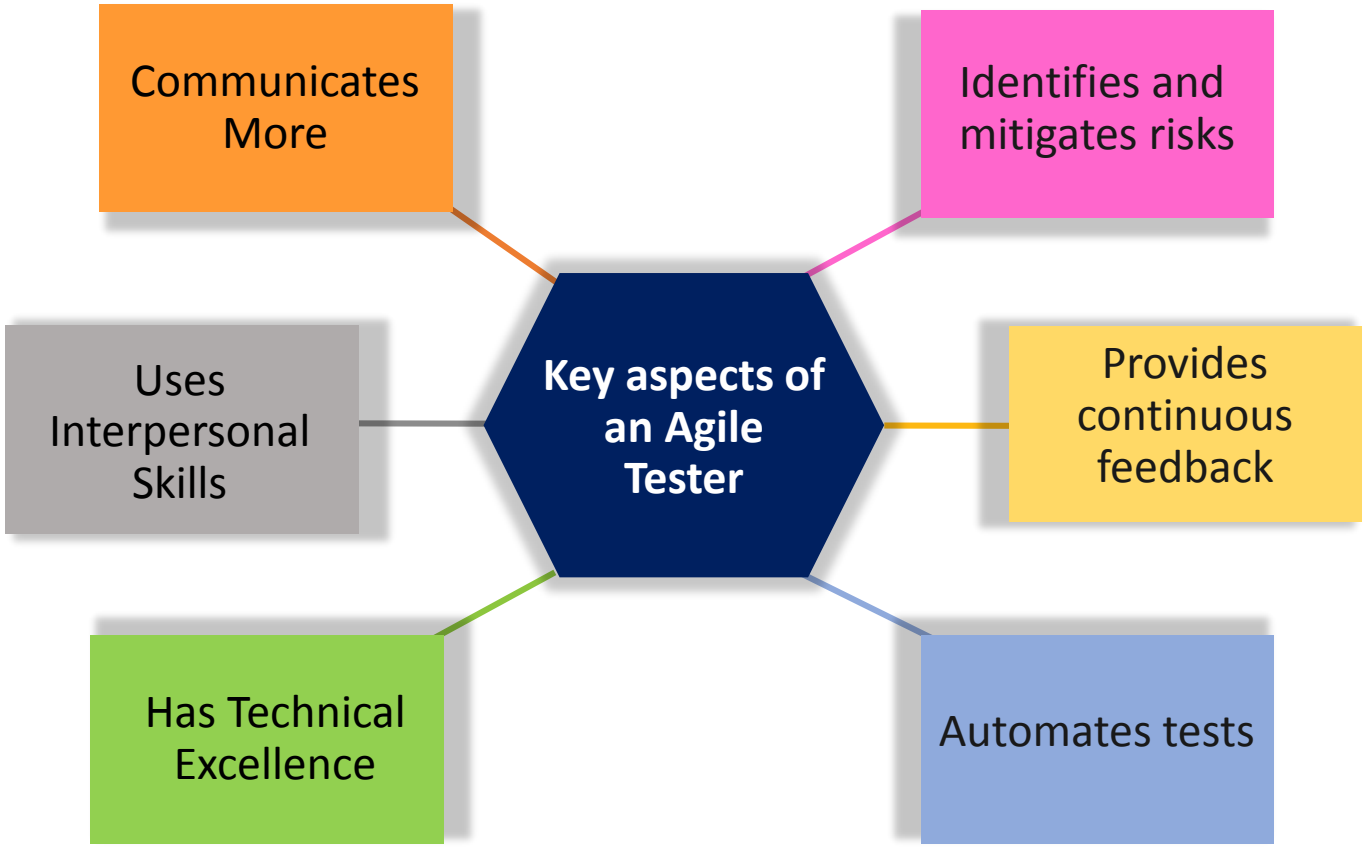
- Ensures the system meets non-functional requirements
- Supported by tools such as load, performance, or specialized tools

© Copyright IBM Corporation 2015

What is the **role** of a Tester in an Agile environment?

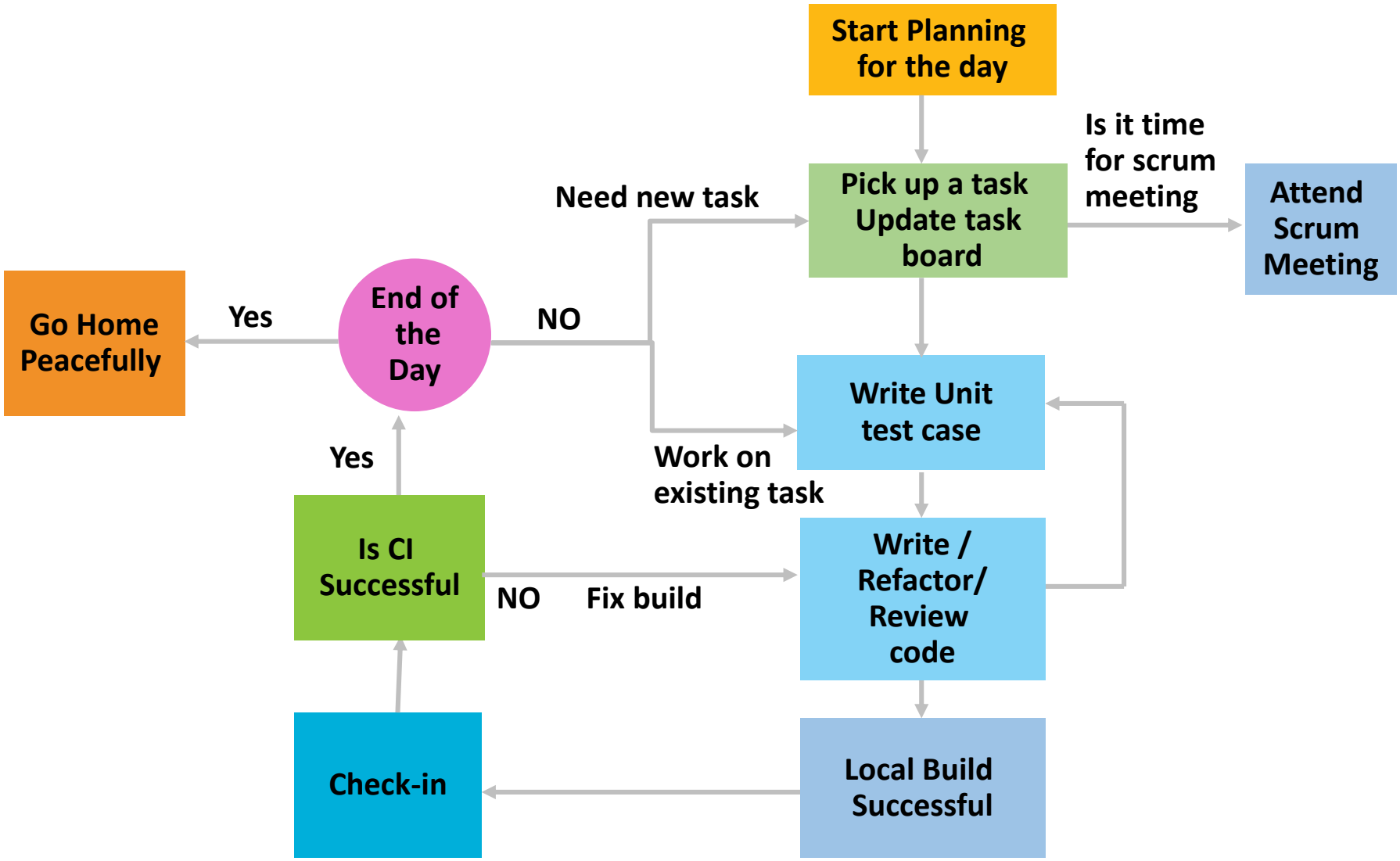
Brighter Blue

Testers are a seamless part of the team to improve and build quality into the system and work closely with the product owner. Following are the key aspects of an agile tester.



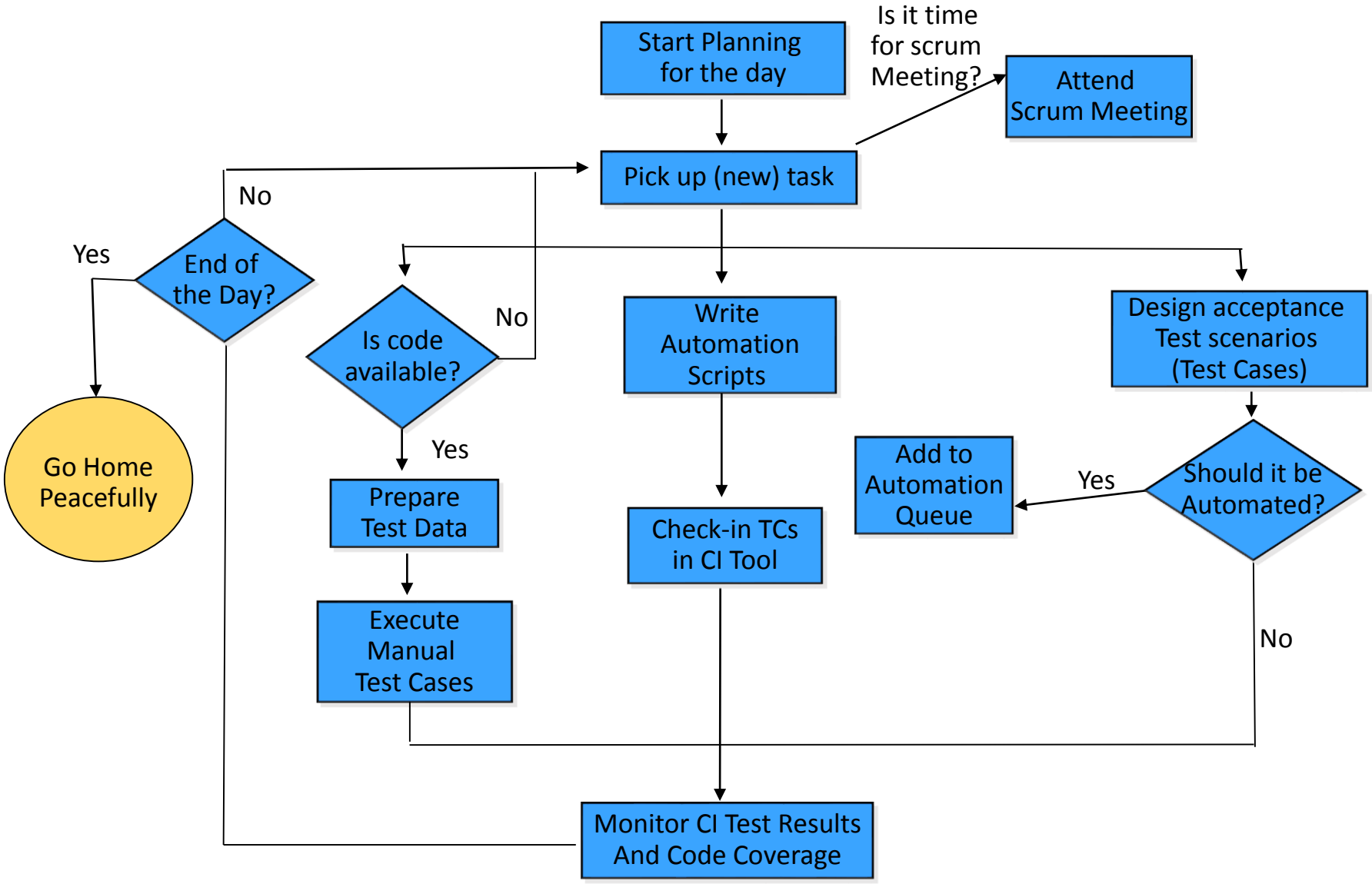
© Copyright IBM Corporation 2015

A typical day in the life of a **developer**



© Copyright IBM Corporation 2015

A typical day in the life of a **tester**



© Copyright IBM Corporation 2015



© Copyright IBM Corporation 2015

Spot Quiz

Brighter Blue

01 Quadrant 2 in the Agile Test Quadrants depicts

A	Unit and Component Tests
B	System-level tests
C	Non-functional tests
D	User facing story and functional tests

© Copyright IBM Corporation 2015